

Everything you should know (a little bit at least) and will be applied (eventually):
Mathematics Training (Basic Knowledge)

Number Theory

Terms: Decimal Representations of Integers. Divisibility, Criteria of divisible by 2, 3, 4, 5, 8, 9, 11, etc. Prime and Composites, G.C.D. (H.C.F.) and L.C.M, Euclidean Algorithm. Odd and Even Number, Parity. Unique Factorizations of Integers, Sum and Number of Factors. Perfect Squares. Modulo Arithmetic. Congruence, Diophantine Equations. Arithmetic Functions, Gauss's Function (Integer Part and Fractional Part). Primitive Roots, Law of Quadratic Reciprocity.

Theorems: Chinese Remainder Theorem. Euclidean Algorithm. Fermat Little Theorem, Euler's Theorem. Law of Quadratic Reciprocity. Beatty's theorem. Wohstemholme. Cauchy-Davenport. Erdos Ginsberg, Chevally's theorem of Finite fields. Legendre's theorem (p-adic valuation). Lifting exponent lemma.

Methods: Infinite Descent.

Algebra

Coordinization, Quantification, Measure, Structure, Representation.

Terms: Polynomials, Synthetic Divisions, Remainder Theorem, Viète's Formula (Vieta flip). Partial Fractions. Standard Identities. Methods of Factorizations: Adding Terms, Completing Squares, Undetermined Coefficients. Quadratic Equation, Sums and Products of Roots. Complex Numbers, Representations, DeMoivre's Theorem, Euler's Formula, Applications to Geometry. Sequences. Functional Equations (Cauchy, D'lambert).. Cyclotomic Polynomials.

Vectors, Matrices, Group, Ring, Field, Finite Field.

Theorems: Remainder Theorem, DeMoivre's Theorem, Euler's Formula, Descartes rule's of signs.

Inequalities

Terms: Arithmetic-Geometric Inequality, Cauchy-Schwarz Inequality, Re-Arrangement Inequalities, Power Mean Inequality, Holder's Inequality, Minkowski's Inequality, Chebyshev's Inequality, Muirhead's Inequality. Convex Functions, Jensen's Inequality. Abel Inequality.

Geometry

Terms: Basic Triangle and Circle Properties and Inequalities. Centers of a Triangle (Incenter, Circumcentre, Centroid, Orthocenter). Power of a point with respect to a circle, radical axis and centre (intersecting chord theorem). Menelaus, Ceva, Ptolemy, Simson

Theorems. Trigonometry applied to Geometry, Cosine Law, Sine Law. Coordinate Geometry applied to Geometry. Complex Numbers. Concurrence of Lines and Points. Weitzenbeck Inequalities, Erdos-Moedell Inequality. Euler Line. Symmedian.

Theorems: Menelaus, Ceva, Ptolemy, Simson Theorems. Cosine Law, Sine Law. Weitzenbeck Inequalities, Erdos-Moedell Inequality.

Transformations: Homothety, Isogonal Conjugate, Inversions. Reflection. Translation. Transformations.

Combinatorics

Terms: Enumerations. Sets and Sunsets. Convex Hull. Colorings. Pigeon-Hole Principle. Graph Theory (degree, connectivity, complete graphs etc.). Trees, Eulerian and Hamiltonian Graphs, Planar graphs (Euler's formula). Addition and Multiplication Principles. Ramsey Theory. Generating Functions.

General Principles

Pigeon-Hole Principle. Inclusion-Exclusion Principle. Extremal Principles. Partition, Covering. Induction. Infinite Descent. Proof by Contradiction. Methods of Exhaustions. Maximum, Minimum, Average Principle.

Mathematics Training (Techniques of Problem Solving)

1. Basic Features:
 - Problem Solving in a context: engineers, entrepreneurs, mathematicians, all have their own problems, we should concentrate on our subject.
 - Non-routine, more than 1 step.
 - Basic ingredient and logic cannot be ignored, subject content.
 - Learning problem solving by solving problems, not a common habit in HK, Proof not emphasized?
2. Why we want to solve problems?
 - Problems are there
 - Enhance understanding, e.g. (FLT).
 - Train mind, help us to solve problems in other fields, rationalize, approximate, exact, complexity.
3. Usual Steps taken in trying to solve a problem:
 - Analysis and design.
 - Exploration.
 - Verification.
 - Polya: *How to solve it*, heuristic strategy, move forward, move backward, employed techniques appropriately.
4. Analysis and Design
 - Available data.
 - What is given, why certain conditions are given? Look for Main Ingredients.
 - Mathematical context.
 - Strategy: draw a diagram, specific values, etc.
5. Exploration
 - Equivalent problems, contradiction.
 - Modify the problem.
 - Patterns.
6. Verifications
 - All data used?
 - Conform to reasonable estimates?
 - Can it be obtained differently?
 - Relate to something elsewhere?
 - Suitable Models?

Readings

Mathematical Excalibur, Art of Problem Solving (Aops), Mathematical Database, etc.

Contests: *USAMOs, Russian Mathematical Olympiads, Hungarian Problem Books, Chinese Mathematical Olympiads, Romanian Mathematical Olympiads, etc.*

Suggestions:

- (i) Follow our program rigorously.
- (ii) Read intensively other Contest Papers yourself.
- (iii) Strong basic Understanding of the materials.
- (iv) Focus.
- (v) Know how much you know but don't lose your confidence as compared with others.
You know as much as anybody.
- (vi) Improve both qualitatively and quantitatively, quickly and significantly).
- (vii) At the end if we can stay depends on how our new recruits do. (We may choose a smaller team).
- (viii) Every year starts anew.
- (ix) Girls: be more audacious.

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